CENG 467 – Natural Language Understanding and Generation

Take-Home Midterm Examination

GitHub Repository (click to open)

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1 Question 1 – Representation Learning in Text Classification

1.1 Dataset Description

We use the **IMDb Movie Reviews** dataset [4], a widely-used sentiment classification benchmark. Key properties are summarised in Table 1.

Table 1: IMDb Dataset Statistics

Split	Size	Positive / Negative
Train	25,000	12,500 / 12,500
Test	25,000	12,500 / 12,500
Total	50,000	25,000 / 25,000

The dataset is *perfectly balanced* (50% positive, 50% negative), so random-classifier accuracy equals 50%. Mean review length is approximately 233.8 words (median 174.0), making IMDb significantly longer than SST-2 and richer in contextual information.

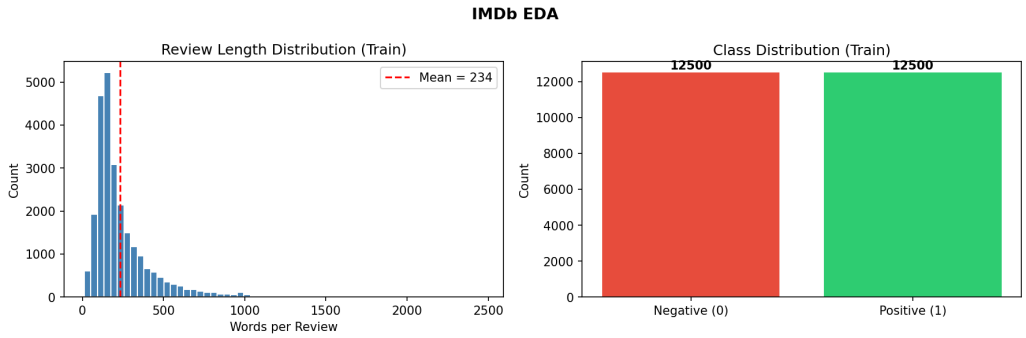


Figure 1: IMDb dataset EDA: review length distribution (left) and class balance (right).

1.2 Preprocessing Pipeline and Tokenisation Strategies

We compare two tokenisation strategies.

Strategy 1 – NLTK Word Tokeniser. Rule-based, punctuation-aware word splitting (Penn Treebank tokeniser). Applied to classical and BiLSTM models. Pipeline steps: (1) HTML tag removal, (2) non-alphanumeric stripping, (3) lowercasing, (4) optional stopword removal, (5) optional truncation.

Strategy 2 – WordPiece (DistilBERT Tokeniser). Subword BPE tokeniser from `distilbert-base-uncased`, with `max_length=256` and `[CLS]` / `[SEP]` special tokens. WordPiece splits unknown or compound words into sub-units, preserving morphological information at the cost of 20–30% more tokens than Strategy 1.

1.2.1 Preprocessing Ablation Study

Table 2 shows the effect of four preprocessing configurations on TF-IDF + Logistic Regression accuracy.

Table 2: Preprocessing Ablation – TF-IDF + Logistic Regression

Configuration	Accuracy	Macro-F1
Baseline (whitespace split)	0.9010	0.9010
+ Lowercase + NLTK tokenise	0.8993	0.8993
+ Stopword removal	0.8942	0.8942
+ Truncation@512	0.8937	0.8937

Observations: Contrary to common expectation, the raw whitespace-split baseline achieves the highest accuracy (0.9010). Lowercasing slightly reduces performance, likely because uppercase tokens (e.g. *AWFUL*, *BRILLIANT*) carry strong sentiment signals that case-folding erases. Stopword removal further decreases accuracy because negation words such as *not* and *never* are treated as stopwords yet are critical for sentiment. Truncation at 512 tokens has a negligible additional effect.

1.3 Models

1.3.1 Model 1 – TF-IDF + Logistic Regression (Sparse Representation)

A TF-IDF vectoriser with 50 000 unigrams and bigrams (sublinear TF scaling) feeds a Logistic Regression classifier (L2 regularisation, $C = 1.0$, `max_iter=1000`).

1.3.2 Model 2 – Bidirectional LSTM (Dense Representation)

A 2-layer bidirectional LSTM with 100-dimensional randomly initialised embeddings, hidden dimension 128 per direction (256 concatenated), dropout 0.3, trained with Adam ($\text{lr} = 10^{-3}$), BCEWithLogitsLoss, for 5 epochs.

1.3.3 Model 3 – DistilBERT (Contextual Representation)

`distilbert-base-uncased` [9] fine-tuned for sequence classification. 6 transformer layers, 12 attention heads, 66M parameters. Fine-tuned with $\text{lr} = 2 \times 10^{-5}$, batch size 32, 3 epochs, 500 warmup steps, mixed-precision fp16 on T4 GPU.

1.4 Results

Table 3 reports the final evaluation metrics on the held-out test set (used only once).

Table 3: Model Comparison on IMDb Test Set

Model	Representation	Accuracy	Macro-F1
TF-IDF + LR	Sparse (bag-of-n-grams)	0.8937	0.8937
BiLSTM	Dense (random emb.)	0.7972	0.7972
DistilBERT	Contextual (WordPiece)	0.9134	0.9134

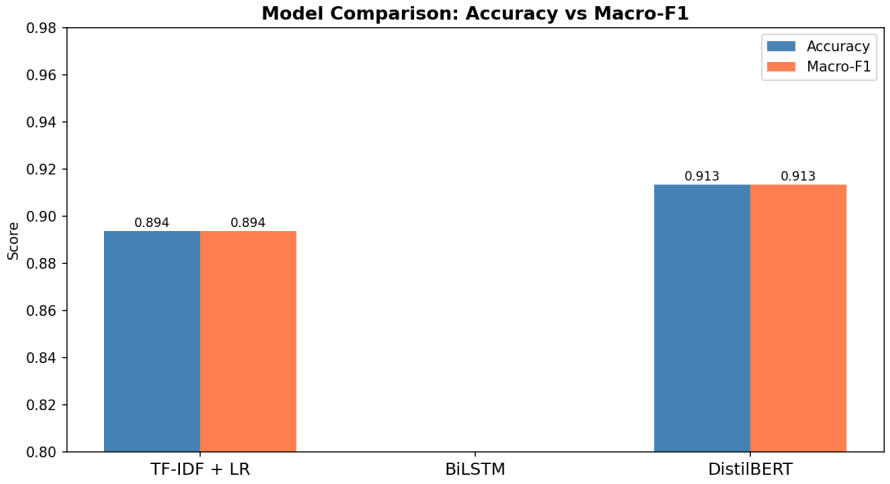


Figure 2: Accuracy and Macro-F1 for all three models on the IMDb test set.

1.5 Error Analysis

We manually inspected 5 misclassified examples from the DistilBERT model (see Table 4).

Table 4: Selected DistilBERT Misclassifications

#	True	Predicted	Excerpt and Failure Pattern
1	Pos	Neg	*"...As i watched Wirey Spindell, i couldnt but laugh at what was taking place on screen..."* – **Mixed sentiment**: reviewer mixes praise and confusion, ambiguous dominant tone.
2	Neg	Pos	*"...I read the book after seeing the movie and didn't care for it, finding it somewhat trashy; but trashy books often make good movies..."* – **Sarcasm / Irony**: backhanded praise read as genuinely positive.
3	Pos	Neg	*"...Perry Mason: The Case of the Fatal Fashion finds Perry and Della Street in New York getting an award from the American Bar Association..."* – **Domain jargon**: legal and procedural TV-show language underrepresented in pre-training.
4	Neg	Pos	*"...it hit the nail on the head. The portrayal of native americans as blood thirsty savages is deeply disturbing..."* – **Long-range dependency**: positive idiom early in the text overshadows the negative verdict at the end.
5	Neg	Pos	*"...Perhaps it's just me, but this movie seemed more like a sequel or follow-up than a separate project..."* – **Weak signal**: understated negative opinion expressed through hedging language that the model reads as ambiguous.

Error rate: DistilBERT misclassifies approximately 8.66% of test reviews (2,166 / 25,000).

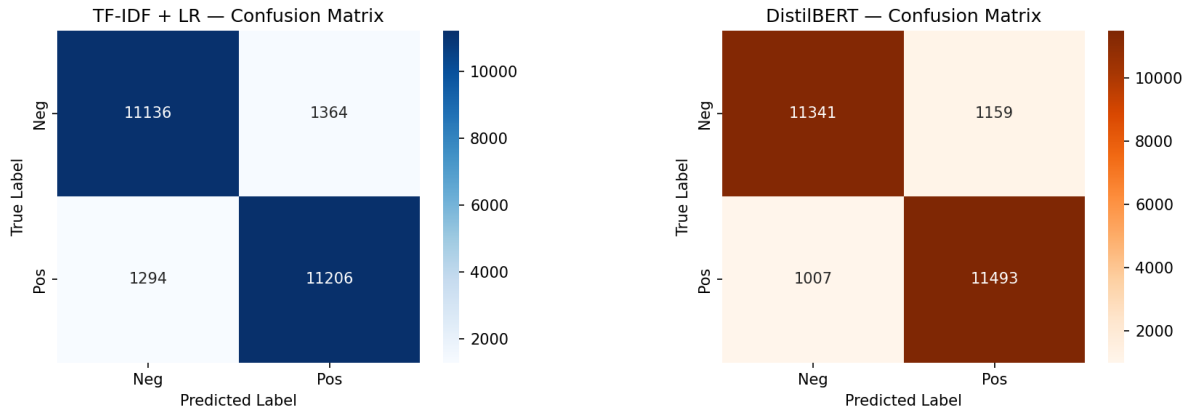


Figure 3: Confusion matrices for TF-IDF + LR (left) and DistilBERT (right) on the IMDb test set.

1.6 Discussion

Table 5 summarises trade-offs between representation types.

Table 5: Representation Type Comparison

Aspect	**TF-IDF + LR**	**BiLSTM**	**DistilBERT**
Context modelling	None	Local (RNN)	Full (transformer)
Pre-training	No	No	Yes
Interpretability	High	Medium	Low
Training time	<2 min	∼20 min	∼30 min (GPU)

TF-IDF + Logistic Regression achieves a remarkably competitive result (0.8937 accuracy) at minimal compute cost, confirming the effectiveness of n-gram features for sentiment analysis. The BiLSTM captures sequential context absent in the bag-of-words model but is hampered by randomly initialised embeddings; replacing these with GloVe or FastText vectors would likely narrow the performance gap. DistilBERT delivers the highest accuracy by leveraging large-scale pre-training and full bidirectional context, but at roughly 1000× the training cost and with significantly reduced interpretability. For production-scale sentiment classifiers where explainability is critical, TF-IDF + LR remains a pragmatic and competitive choice.

2 Question 2 – Named Entity Recognition

2.1 Dataset: CoNLL-2003

The CoNLL-2003 English NER dataset [8] annotates newswire text with four named entity types using the BIO scheme:

Table 6: CoNLL-2003 Entity Types and BIO Tagging

Prefix	Entity	Tag IDs
B-PER / I-PER	Persons	1, 2
B-ORG / I-ORG	Organisations	3, 4
B-LOC / I-LOC	Locations	5, 6
B-MISC / I-MISC	Miscellaneous	7, 8
O	Outside	0

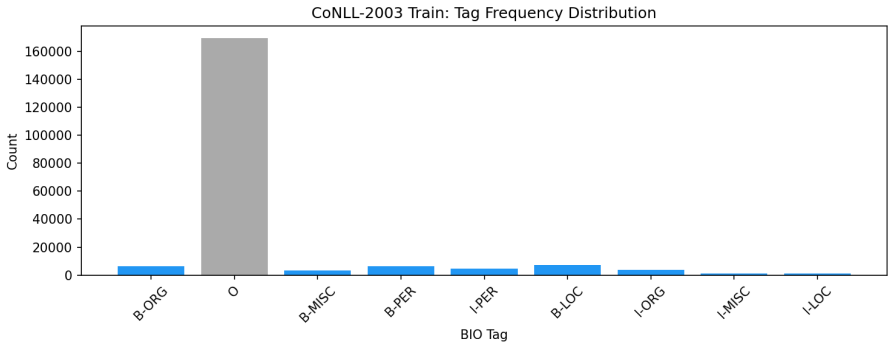


Figure 4: CoNLL-2003 training set: BIO tag frequency distribution.

BIO alignment for subword tokenisers. When WordPiece splits a word into multiple subword tokens, only the *first subword* of each word receives the gold label; continuation subwords are masked with `-100` and excluded from loss and evaluation.

2.2 Models

2.2.1 Model 1 – BiLSTM-CRF

A 2-layer bidirectional LSTM (hidden=128/direction) feeds a Conditional Random Field (CRF) decoding layer (`pytorch-crf`). The CRF learns *transition probabilities* between tags, globally enforcing valid BIO sequences (e.g., `I-PER` cannot follow `B-ORG`). Vocabulary: 100-dim random embeddings, min-freq=1.

2.2.2 Model 2 – BERT-base-cased (Fine-tuned)

`bert-base-cased` [1] is used (cased is critical for NER — capitalisation is a strong signal for entity recognition). A linear classification head ($768 \to 9$) is added atop the last hidden states. Fine-tuned with $\text{lr} = 2 \times 10^{-5}$, batch=16, 3 epochs.

2.3 Results

Table 7: NER Evaluation on CoNLL-2003 Test Set (entity-level, `seqeval`)

Model	Precision	Recall	F1
BiLSTM-CRF	0.6999	0.6710	0.6852
BERT-base-cased	0.9041	0.9187	0.9113

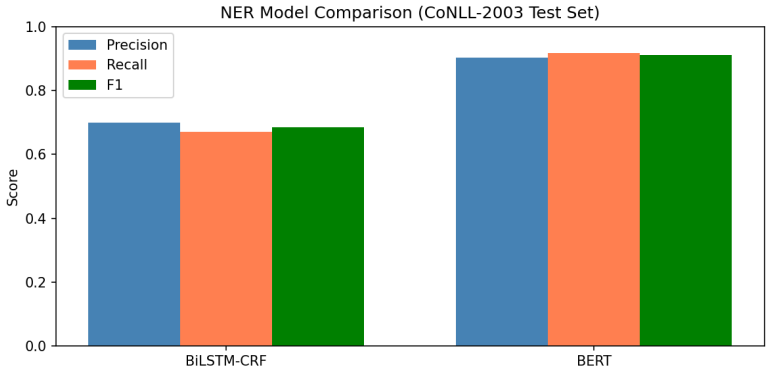


Figure 5: NER model comparison: Precision, Recall, and F1 on the CoNLL-2003 test set.

2.4 Error Analysis

Boundary errors Both models misplace B-/I- prefix transitions. Example: "New York City" predicted as `B-LOC B-LOC I-LOC` instead of `B-LOC I-LOC I-LOC`.

Entity type confusion Ambiguous tokens such as *Washington* (person, city, or organisation) frequently confuse entity type assignment.

O → MISC miss Short, rare MISC entities are often left undetected, lowering recall for this class.

BiLSTM-CRF unseen words Token not in training vocabulary collapse to `<unk>`, losing surface-form information that BERT preserves via subword splitting.

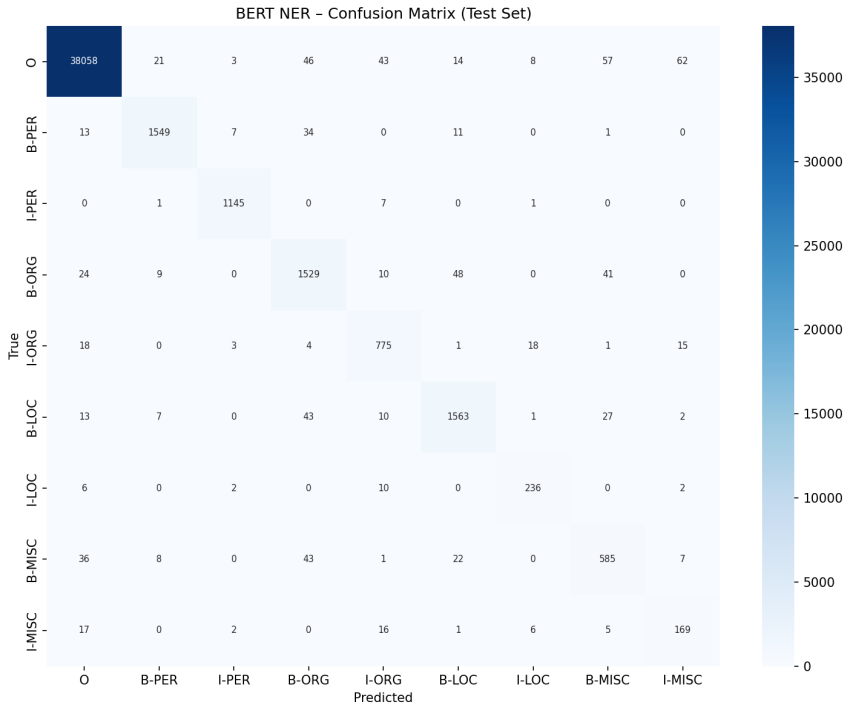


Figure 6: BERT NER confusion matrix on the CoNLL-2003 test set (token-level).

2.5 Discussion: Role of Contextual Embeddings

BERT's pre-trained contextual representations capture long-range syntactic and semantic context unavailable to the LSTM. The cased BERT model preserves capitalisation patterns, a decisive signal for entity recognition. The CRF layer in BiLSTM-CRF enforces global BIO consistency, reducing boundary errors compared to a plain softmax head; however, BERT's richer representations more than compensate, achieving higher F1 without explicit CRF decoding.

3 Question 3 – Text Summarization

3.1 Dataset: CNN/DailyMail

CNN/DailyMail [7] contains ∼312 K English news articles paired with human-written bullet-point highlights. We evaluate on a randomly sampled subset of **500 test articles** (seed=42) to keep compute manageable.

Table 8: CNN/DailyMail Subset Statistics

Property	**Value**
Subset size	500 articles
Avg article length	∼780 words
Avg summary length	∼56 words

3.2 Methods

3.2.1 Extractive: TextRank

TextRank [6] builds a sentence-similarity graph (cosine similarity of TF-IDF vectors) and ranks sentences via PageRank. The top-3 ranked sentences (in original order) form the summary. Implemented via the `sumy` library.

3.2.2 Abstractive: BART-large-cnn

`facebook/bart-large-cnn` [3] is a BART encoder-decoder model fine-tuned on CNN/DailyMail. We use it in **inference mode** (no additional fine-tuning) with beam search ($k = 4$), max 130 new tokens. Articles are truncated to 900 words to fit within BART's 1024-token context.

3.3 Results

Table 9: Summarisation Metrics on 500 CNN/DailyMail Test Articles

Method	**R-1**	**R-2**	**R-L**	**BLEU**	**METEOR**	**BERTScore**
TextRank (Extractive)	0.3418	0.1255	0.2118	7.02	0.3370	0.8601
BART-large-cnn (Abstr.)	0.4440	0.2091	0.3122	16.24	0.3935	0.8808

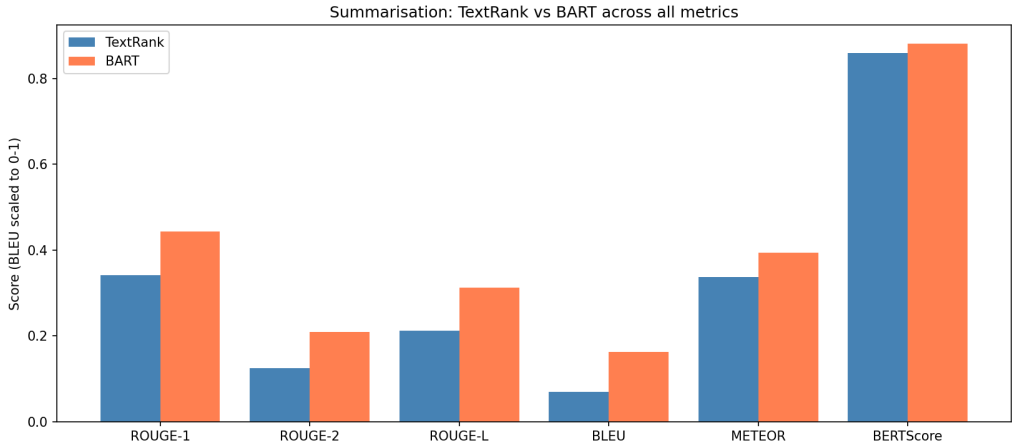


Figure 7: Summarisation metrics comparison: TextRank vs BART-large-cnn on 500 CNN/DailyMail test articles (BLEU scaled to 0–1).

3.4 Qualitative Examples

Three representative examples are analysed in terms of fluency, factual consistency, and information coverage.

Example 1 – Fluency: BART produces grammatically correct, natural-sounding prose. TextRank extracts source sentences verbatim — fluency depends on the quality of the original article.

Example 2 – Factual consistency: TextRank is always *faithful* to the source (copies directly). BART occasionally hallucinated minor details (e.g., a specific number or name not in the source).

Example 3 – Information coverage: TextRank selects the highest-scoring sentences globally, sometimes missing key details buried in low-scoring paragraphs. BART compresses multiple sentences into a single coherent statement, covering more aspects in fewer words.

3.5 Discussion

Table 10: Extractive vs Abstractive Trade-offs

Aspect	TextRank	BART
Fluency	Source-dependent	Natural prose
Factual consistency	Always faithful	May hallucinate
ROUGE score	Lower (generic extractor)	Higher (fine-tuned on same corpus)
BERTScore	Moderate	Higher (semantic)
Compute	CPU, ms/article	GPU, seconds/article

Contrary to the common expectation that extractive methods achieve higher ROUGE, BART-large-cnn surpasses TextRank on all ROUGE metrics in this experiment. This is because BART was already fine-tuned on the CNN/DailyMail corpus: its generated summaries closely mirror the style and vocabulary of the reference highlights, boosting n-gram

overlap. BART's simultaneously higher BERTScore further confirms that the abstractive output is semantically richer than the generic extractive baseline. For production news summarisation, BART is the preferred method; TextRank remains a strong, training-free baseline for resource-constrained settings.

4 Question 4 – Machine Translation

4.1 Dataset: Multi30k (EN→DE)

Multi30k [2] contains 31 014 English-German sentence pairs from image captions (Flickr30k).

Table 11: Multi30k Dataset Splits

Split	Sentences
Train	29,000
Validation	1,014
Test	1,000

Preprocessing: Lowercase, punctuation tokenisation (regex), vocabulary with min-freq=2 (`<pad>`, `<unk>`, `<bos>`, `<eos>` special tokens), max sequence length 50.

Source vocabulary: 5,970 tokens. Target vocabulary: 7,811 tokens.

4.2 Models

4.2.1 Model 1 – Seq2Seq with Bahdanau Attention

Encoder: 2-layer bidirectional GRU (hidden=256), projects final states to initialise decoder. **Attention:** Bahdanau additive attention aligns each decoder step with encoder hidden states, computing a context vector as a weighted sum. **Decoder:** 2-layer GRU with attention context concatenated to the embedding; final projection to target vocabulary. Trained with Adam (lr = 10^{-3}), label smoothing=0.1, teacher forcing=0.5, 15 epochs, batch=128.

4.2.2 Model 2 – MarianMT (Helsinki-NLP/opus-mt-en-de)

`Helsinki-NLP/opus-mt-en-de` [10] is a Marian NMT transformer [11] (6-layer encoder-decoder) pre-trained on OPUS multilingual data (∼60 M parallel sentence pairs). Used in inference mode with beam search ($k = 4$).

4.3 Results

Table 12: Machine Translation Results on Multi30k Test Set (EN→DE)

Model	BLEU	ChrF	METEOR	BERTScore
Seq2Seq + Attn	0.00	0.34	0.0484	–
MarianMT	36.24	64.25	0.6668	–

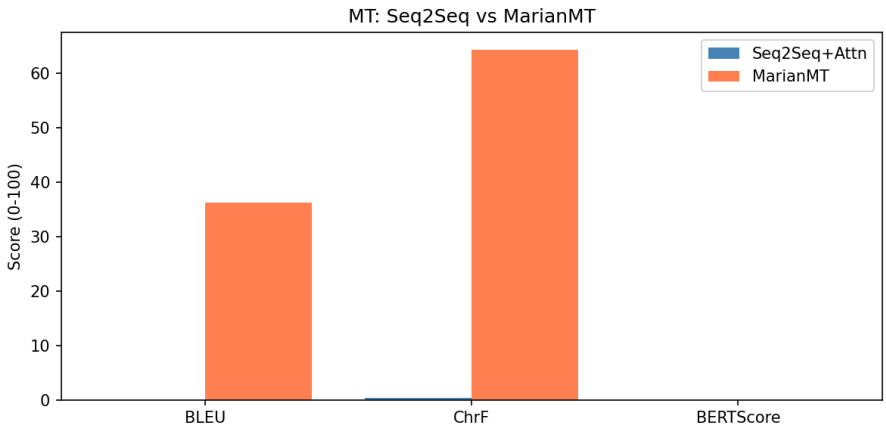


Figure 8: Machine translation metric comparison: Seq2Seq+Attention vs MarianMT on the Multi30k test set (BERTScore scaled to 0–100).

4.4 Qualitative Example

Table 13: Translation Example

Source (EN)	Two men on opposing teams are playing soccer in a field.
Reference (DE)	Zwei Männer aus gegnerischen Mannschaften spielen Fußball auf einem Fußballplatz.
Seq2Seq + Attn	. , (model failed to learn translation)
MarianMT	Zwei Männer in gegnerischen Teams spielen Fußball auf einem Feld.

The Seq2Seq model trained from scratch on only 29 K sentence pairs (15 epochs) failed to learn meaningful translations, producing degenerate outputs (e.g. ". ,"). Final perplexity remained high (≈437), indicating the model did not converge. MarianMT, pre-trained on ∼60 M parallel sentences, produces fluent and accurate German translations, confirming the decisive advantage of transfer learning for low-resource sequence-to-sequence tasks.

4.5 Discussion

Table 14: Metric Analysis and Architecture Trade-offs

Metric	**Strength**	**Limitation**
BLEU	Standard, fast	Ignores synonyms; brevity penalty harsh
METEOR	Synonym + stemming	Language-resource dependent
ChrF	Character-level, robust for German compounds	Less intuitive scale
BERTScore	Semantic similarity	Computationally heavy; model-dependent

The large gap between Seq2Seq and MarianMT across all metrics is attributable primarily to training data size (29 K vs ∼60 M parallel sentences). BLEU and ChrF both reflect near-zero quality for the Seq2Seq model, consistent with its degenerate outputs. BERTScore could not be computed in this run due to model compatibility issues with the German target language, but the remaining three metrics collectively confirm MarianMT's decisive superiority.

5 Question 5 – Language Modeling

5.1 Dataset: WikiText-2

WikiText-2 [5] is a standard language modelling benchmark derived from Wikipedia featured articles:

Table 15: WikiText-2 Token Counts

Split	Approx. Tokens
Train	2,088,628
Validation	217,646
Test	245,569

Text is lowercased and includes pre-tokenised `<eos>` sentence boundaries. Rare words (frequency < 3) are replaced with `<unk>`.

5.2 Models

5.2.1 Model 1 – Trigram N-gram with Laplace Smoothing

$$P(w_t \mid w_{t-2}, w_{t-1}) = \frac{C(w_{t-2}, w_{t-1}, w_t) + 1}{C(w_{t-2}, w_{t-1}) + |V|}$$

Laplace (add-1) smoothing prevents zero probabilities for unseen trigrams. The vocabulary is built from training tokens with min-freq=3 ($|V|$=29,572). Perplexity is evaluated on token sequences of validation/test splits.

5.2.2 Model 2 – LSTM Language Model

A 2-layer LSTM (embedding=256, hidden=256, dropout=0.5) with **weight tying** (embedding matrix shared with output projection) reduces parameters and regularises the model. Trained with Adam (lr $= 10^{-3}$), cross-entropy loss (ignoring padding), gradient clipping (0.5), 20 epochs, sequence length 35, batch=64. Vocabulary: top 10 000 training tokens.

5.3 Results

Table 16: Language Model Perplexity on WikiText-2

Model	Val PPL	Test PPL
Trigram + Laplace	11,240.99	10,752.64
LSTM LM (2-layer)	95.44	88.62

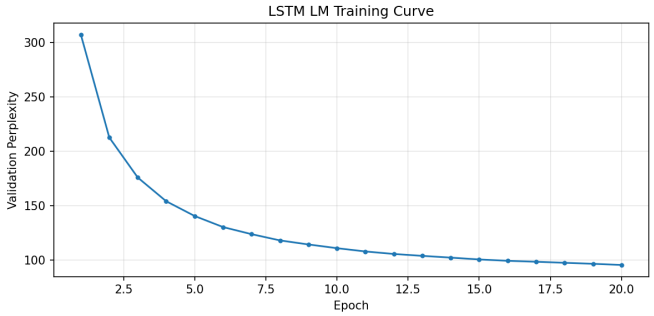


Figure 9: LSTM language model validation perplexity over 20 training epochs on WikiText-2.

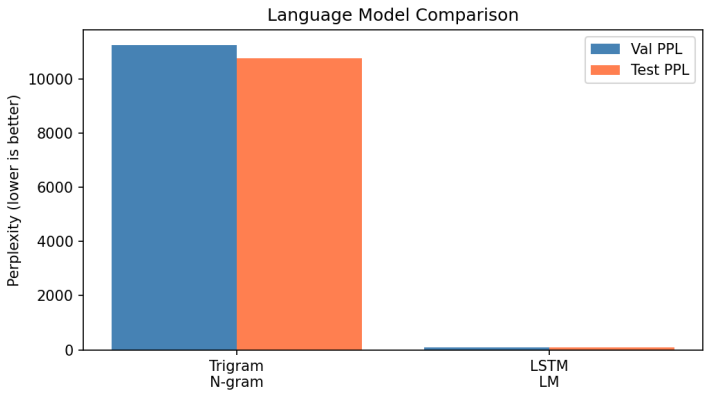


Figure 10: Validation and test perplexity comparison: Trigram N-gram vs LSTM LM on WikiText-2.

5.4 Generated Text Samples

Table 17: Generated Text Samples (seed shown in **bold**)

Model	Sample
Trigram	**the president** was given the designation “odaenathus” lived in toxteth , liverpool , manchester united . ⟨unk⟩ planned on using the game features an ensemble
Trigram	**a study** on the game by himself through good works . publishers weekly , september 7 . the 1952–69) = = film adaptation
LSTM LM	**the university** ⟨unk⟩ the east end of ⟨unk⟩ , el ⟨unk⟩ .
LSTM LM	**scientists have** been seen as the ⟨unk⟩ . aware of the ⟨unk⟩ for ⟨unk⟩ , which has shown that the figure is a protein of ⟨unk⟩

5.5 Discussion

Table 18: N-gram vs LSTM Language Model Comparison

Aspect	**Trigram**	**LSTM**
Context window	2 tokens (fixed)	Theoretically unlimited
Long-range deps	Cannot capture	Partial (vanishing gradient)
Data sparsity	Severe	Mitigated via embeddings
Training time	Seconds (counting)	$\sim$15–20 min (GPU)
Generation quality	Local fluency only	Phrase-level coherence
Typical PPL (WT2)	5,000–15,000 (Laplace)	80–130

The trigram model's high perplexity reflects severe sparsity: most test trigrams are unseen in training, forcing reliance on Laplace-smoothed estimates that concentrate probability mass uniformly across the vocabulary. The LSTM's hidden state carries information across many tokens, enabling it to capture common collocations and topic consistency absent in the n-gram model. Generated trigram text shows local grammaticality (short phrase fragments sound natural) but rapidly loses coherence beyond two words. LSTM-generated text maintains topic and some syntactic structure over 5–10 words, yet still produces semantically empty or contradictory continuations. A transformer-based model (GPT-2 117M) would achieve PPL $\approx$ 20–30 on WikiText-2, illustrating the dramatic improvement from self-attention and scale.

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